

Pediatric Patient Counting with Interactive Object Detection

A Journey Through the Evolution of Occlusion-Resistant Computer Vision

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The Clinical Crisis: The “Invisible Child”

The Reality of Ghanaian OPDs: Infants are culturally carried tightly on their mothers' backs. When they enter the hospital, electronic health records (like LHIMS) systematically register the adult, but often miss the child.

The Data Deficit

When a significant percentage of pediatric patients is rendered invisible to administrative data, hospitals suffer from artificial resource droughts. Budgets for pediatric therapeutics, bed space, and vaccines are dangerously under-allocated.

The Goal: Automate the counting process using standard hospital cameras.

The Reality: Standard “off-the-shelf” AI fails completely in this environment.

The Intellectual Journey

Solving this crisis required deconstructing standard Artificial Intelligence and returning to fundamental mathematics. We will walk through the three major evolutionary bottlenecks:

1. **The Classification Journey:** How do we tell a child from an adult?
2. **The Occlusion Journey:** Why do algorithms literally erase carried babies?
3. **The Tracking Journey:** How do we remember a child when they disappear from view?

Phase 1: The Classification Journey

Era 1: Deep Learning & Age Regression (The Standard Approach)

Current state-of-the-art AI uses Convolutional Neural Networks (CNNs) to extract high-dimensional facial geometry (wrinkles, skin texture) to guess a person's age.

Why it failed in our hospitals:

- **Dataset Bias:** Models trained on Western datasets (like COCO) expect adults in standard, upright poses.
- **Hardware Limits:** CNNs require massive GPU power (FLOPS) unavailable in rural clinics.
- **Visual Limits:** Ceiling cameras look down at the tops of heads; facial textures are invisible.

Phase 1: The Mathematical Pivot

Era 2: Geometric Models & Allometry (Our Approach)

Instead of relying on fragile pixel patterns, we pivot to mathematical biology. Human growth follows non-linear power laws (Allometric Scaling).

The Head-to-Body Canon (C)

By extracting just two bounding boxes, we calculate a rigid biological constant:

$$C = \frac{\text{Body Height } (I_B)}{\text{Head Length } (I_H)}$$

- **Toddler:** $C \in [5.3, 6.1]$ **Adult:** $C \geq 7.5$
- **The Breakthrough:** This simple vector division is computationally “cheap” and perfectly scale-invariant. It works regardless of camera quality.

Phase 2: The Severe Occlusion Journey

Once we can identify a child, we must find them when they are tied to a mother's back.

Era 1: Bounding Boxes & Hard NMS (The Standard Approach) Standard AI draws boxes around humans. To avoid double-counting the same person, it uses Non-Maximum Suppression (NMS) based on the Intersection over Union (IoU):

$$\text{IoU}(A, B) = \frac{A \cap B}{A \cup B}$$

The Fatal Flaw: Because the baby's box (B) is entirely inside the mother's box (A), the IoU approaches 1.0. The math assumes this is a mistake and **permanently deletes the baby's box**. Standard AI actively erases our patients.

Phase 2: Overcoming NMS

Era 2: SoftNMS and Skeletal Separation

To stop the AI from erasing the baby, the computer vision community developed two solutions:

1. **SoftNMS:** Instead of a rigid threshold that deletes high-IoU boxes, a Gaussian penalty function gently lowers the confidence score. The baby's box survives.
2. **Part Affinity Fields (PAFs):** We bypass boxes entirely and track joints (skeletons).

Biomechanical Dissonance

The algorithm tries to connect a mother's shoulder to the baby's elbow. The spatial distance and angles violate human biology. The math forces the skeleton to split in two, proving two humans occupy the same space.

Phase 3: The Tracking Evolution (Avoiding Duplicates)

If a mother turns around, the baby is hidden. If the AI loses the baby's ID, it will count them as a “new” patient when the mother turns back.

Era 1: Convolutional Bounding Boxes (Ouyang, 2015)

- Tracked semantic features ($F_l \in \mathbb{R}^{H \times W \times C}$).
- **Flaw:** When occlusion occurs, the baby's pixel features physically merge with the mother's. The identity is lost.

Phase 3: Tracking Eras (Cont.)

Era 2: Anchor-Free Centers (FairMOT, 2021)

- Shifted to tracking just the exact center pixel of a person using a Gaussian heatmap:
$$Y_{x,y,c} = \exp\left(-\frac{(x-x_c)^2 + (y-y_c)^2}{2\sigma_c^2}\right).$$
- **Flaw:** If the baby is completely hidden by the mother's torso, the center-point vanishes.

Era 3: Spatio-Temporal Kinematics (PF-Track, 2023)

- **The Fix:** The Kalman Filter. It uses an 8-dimensional state vector $[u, v, \gamma, h, \dot{u}, \dot{v}, \dot{\gamma}, \dot{h}]^T$ to predict where the hidden object is moving.
- **The Remaining Flaw:** It remembers an object is there, but cannot mathematically prove if the hidden object is a child, or just a large backpack.

The Synthesis: "One Body, Two Identities"

Era 4: Our Proposed Framework (2026)

We bridge the Anthropometric math (Phase 1) with Kinematic memory (Phase 3).

1. **Detect:** Isolate independent Head and Body features.
2. **Trigger:** If one large Body bounds two distinct Heads (h_1, h_2), evaluate the spatial displacement.
3. **Verify (The Canon):** Apply the Allometric Canon ($C \in [4.0, 6.0]$) to the secondary head. This mathematically overrides height biases and proves the "backpack" is actually a pediatric patient.
4. **Track:** Assign anchor-free IDs (FairMOT) and cache the trajectory vectors (Kalman/PF-Track).

The Final Layer: Absolute Privacy

Hospitals cannot legally record or save video of patients.

Edge Computing (REVAMP2T Architecture)

- **Zero Storage:** Video is processed in volatile RAM and permanently destroyed in milliseconds.
- **Triplet Loss Encryption:** We extract the texture of the baby's clothing and convert it into a 1×1280 mathematical vector.
- **Result:** A completely abstract, human-uninterpretable string of numbers. We track the numbers, not the faces.

Bridging the Data Gap

- Standard deep learning was not built for the cultural realities of Ghanaian healthcare.
- By tracing the failures of standard bounding boxes and NMS algorithms, we proved that throwing more computing power at the problem is ineffective.
- By returning to **Mathematical Biology** (Allometry) and **Kinematic Prediction** (Kalman), we engineered an elegant, edge-computing solution.
- The “Invisible Child” becomes actionable, private telemetry, allowing hospitals to accurately allocate life-saving resources.

Thank You.
Questions?

Selected References

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